

Engagement Mechanisms in Workshops for Sustainable Community Development

Bruce A. HECHT ^{a,1}, Ted HALSEY ^b, Mai NGUYEN ^a, Bryan R. MOSER ^{a,c}

^aMassachusetts Institute of Technology, Cambridge, USA

^bHalsey Architecture + Planning, Denver, USA

^cUniversity of Tokyo, Japan

Abstract. Sustainable development for cities and communities presents challenges planning for economic and environmental resiliency. Workshops can bring stakeholders' local knowledge in combination with planning experts' professional knowledge. This paper proposes a system model for workshops operating as transdisciplinary problem-solving systems of functions, processes, and roles. The identified mechanisms link to three steps: first, stakeholders define the relevant problems; second, stakeholders and planning experts contribute to exploring a range of possible decisions. Finally, decision methods illuminate and prioritize options for selecting concepts of desired solutions. An approach to examine and test the workshop system model for a site-specific example focuses on land-use at transit stations. The research aims to improve human capacity to address the growing complexity in cities increasing social equity with economic and environmental resilience.

Keywords. Urban Planning, Sustainable Development, Social Equity, Teamwork, Planning Workshops, Resilience, Cities, Transdisciplinary Engineering

Introduction

The challenges for sustainable development in cities are rising with changes in the physical geography, social, and economic aspects for peoples' lives. In a recent report, the United Nations Habitat Assembly (UN-HABITAT) recognized the interaction between housing, climate, urban living, and the drive towards sustainable development combined with economic development [1]. When considering future growth of a city, questions about how people live, work, build, and inhabit cities reflect stakeholders with multiple interests and diverse needs. Transdisciplinary approaches to these questions offer the potential to include experts with a range of professional disciplines, knowledge, and experiences to engage local civic involvement in decision-making. Fokdal et al. define *transdisciplinary challenges* in the context of urban environments, as "the integration of knowledge from various disciplines (i.e. interdisciplinarity) as well as the involvement of civil society and other non-academic actors into the realm of research and practice" [2]. In a 2022 study of public participation in the Netherlands, Ballering identified that local knowledge and lived experience are complementary to experts' knowledge and experience [3].

¹ Corresponding Author, Mail: b_hecht@mit.edu.

Workshops provide collaborative settings where community members and experts may interact through participatory engagement. Examples of workshop use are often tailored to the community groups involved. Bell et al. described the Collaborate! Series for community groups working with the University College London [4], while Amitage et al. highlighted combining knowledge with social learning in support for sustainable change from a transdisciplinary perspective in learning and practice [5].

As effects of climate change contribute to the social, economic, and environmental dimensions of cities, the importance of transformative change practices are increasingly demanded to drive collaboration and effective adaptation [6]. While participatory approaches are often advocated as desirable for development processes, effective practices may benefit from improved understanding and testing. Technical societies advocate for the value of principles for community engagement as a basis for experts to effectively engage with stakeholders [7].

To help address these questions, we propose a model of urban design workshops operating as problem-solving systems. We present possible approaches that build on teamwork studies and transdisciplinary engineering research. We intend to adapt site-specific needs in the experimental approach to test these models.

1. Teamwork Models in Urban Design Workshops

Researchers in the field of transdisciplinary engineering provide a foundation for building and testing teamwork and system models. In considering urban design workshops, this paper outlines how the ideas and methods developed in teamwork research are adapted to a model of workshops as a system.

Urban design workshops provide a time-bounded setting where a city's residents and community members meet with planning experts and professionals to address needs for growth, equity, change, and resilience [8]. The approach to participation includes two distinct types of processes: *evaluative processes*, where participants react to proposed options, and *collaborative processes*, where participants contribute to identifying options and decisions. While collaborative processes are more favorably perceived, results are mixed and dependent on how workshops are organized [9]. Processes for performance of complex problem solving contribute to understanding and interactions of people in teams [10].

1.1. Models and Land Use Decisions

Urban design workshops make use of models to provide information for participants to engage in conversations and to inform decision-making. Basu and Ferreira illustrate how Land Use Transportation Interaction (LUTI) models may provide insight into changes to vehicle traffic, pedestrian access, and neighborhoods [11]. While the 20th century development of cities was heavily driven by experts seeking to maximize automotive utilization with flow models for vehicle traffic, a complementary approach for pedestrians [12] is placed in the context of the idea of walkable cities as an organizing principle [13],[14].

Recently, agent-based models involving activities, transportation, and population models, applying a site-specific approach for demographic changes to a city in Japan for aging populations [15]. The approach provided exploration of decisions and proposed transportation concepts.

In seeking to model decisions related to land-use, technical optimizations are often combined with system methods [16]. Expanded development for “virtual cities” address integration of road, transit, and other urban networks with population models [17]. These models combine long-term and medium-term time modules that enable several years’ perspective for changes in housing, commercial space, and employment patterns.

The inclusion of systems models provides for informed decisions making and potential for improved engagement of stakeholders in the participatory processes. Moser et al. examined decision-making in stakeholder groups with experts and local stakeholders for infrastructure construction projects in Japan and France [18]. Early decisions involving broader stakeholder participation relates to beneficial improvement in outcomes in construction performance measures and in operational performance measures such as mobility, economics, and environmental factors.

1.2. Participatory Processes

In comparing participatory processes, Raynor, Doyon, and Beer identified collaboration planning for addressing the complexity of stakeholder engagement and decision-making [19]. While many researchers investigate individuals’ decision-making, two types of uncertainties are important in the social context: *risk* and *ambiguity* [20]. Probabilistic decisions involving *risk* are those where the outcome is uncertain, but the resulting payoffs are predictably known. In contrast, *ambiguity* is defined for decisions where the outcome is uncertain, and the payoffs are also uncertain. The importance of this view is (1) problems involving uncertainty have interactions between the team’s social behavior and the uncertainties. (2) social questions may not have an optimal or “correct” answer.

1.3. Awareness and Attention

Areas of interest in teamwork studies relate potential mechanisms for engagement with knowledge identification, communication, attention, and interaction between participants. The elements of *situation awareness* and *attention allocation* defined by Endsley:

Definition 1 (*Situation Awareness*). Situation awareness is defined as an understanding of the situation as a whole, forming a basis for decision making. [21]

Definition 2 (*Attention Allocation*). **Attention allocation is defined as the perception and processing of cues and information. For complex systems, effective deployment of attention allocation is critical to decision making and response planning.** [21]

Tan and Moser developed an experimental method using a project planning exercise. Teams participated in a workshop by planning a project to develop an autonomous vehicle. The study measured attention allocation, decisions made, and interactions between participants [10]. Team performance indicators focused on the number of streams for total iterations and by evaluating meaningful iterations in these streams. Alvarez-Pelegrin, Moser, and Sakhrani considered the attention-decision-learning cycles for teamwork in projects. A maritime shipping challenge was simulated using a computer model. Participants evaluated potential concepts as system architectures to meet emissions reduction, schedule, and costs. Phases, surprises, and sequences were reviewed and compared with performance results. Performance was visualized in the tradespace for the selected architectures, for emissions, cargo, fuel, and cost [22]. Manandhar et al. addressed the effect of attention and awareness for achieving

performance in the selected solutions [23]. The expansion of the investigation of sensemaking provided additional measures in experimental teamwork studies. Vazquez, Faruque, and Moser employed sensemaking and instrumented teamwork using the Mars Star City simulation model [24] as described by McDonough [25].

Integration of individual interactions and teamwork benefits from understanding mechanisms to develop a process model for teamwork systems. Kozlowski and Ilgen identified that multi-goal models may be represented by feedback control mechanisms adjusting for individual performance and team performance [26]. Experimental approaches enable exploration of mechanisms that link to process understanding [27].

2. Urban Design Workshop Goals

Woodworth identifies the roles for participation in the development of creative problem-solving ideas and the definition of workshop goals [8]. In considering complex systems and sustainable development goals, Haskins highlights systems understanding for complex systems [24]. The roles of stakeholders raise questions as to how to best define and achieve urban goals:

1. How are needs defined for urban goals?
2. How well are these needs explored in urban design workshop interactions?
3. How equitably are the benefits balanced and understood by the stakeholders?
4. How does shared understanding contribute to the development of consensus in decision-making?

Participatory civic engagement outlines a foundation for understanding fundamental processes and functions for urban design workshops. Researchers of participation tools and processes consider collaboration between experts and members of the public. Ataman and Tuncer reviewed over 25 years' from 1995-2021 and identified five thematic areas as urbanism, community, sustainability, building types [28]. The review categorized four aspects as participation tools, digital tools, representations, and responsive cities. Kopáček reviewed participation in planning processes with a focus on how experts and local participants exchange information on proposals [29].

2.1. Summary of Key Processes for Urban Design Workshops

Figure 1 illustrates the proposed model for urban design workshops as a system. A flow from the inputs at the top of the diagram begins with identifying the participants and the issues to be addressed. The functions of design workshops are developed in connection with stakeholder's needs. Examples of the practices in use are published in handbooks such as the U.S. National Institute of Building Sciences (NIBS) and the U.S. National Renewable Energy Laboratory (NREL) guidelines for preparing and conducting workshops. [30], [31]

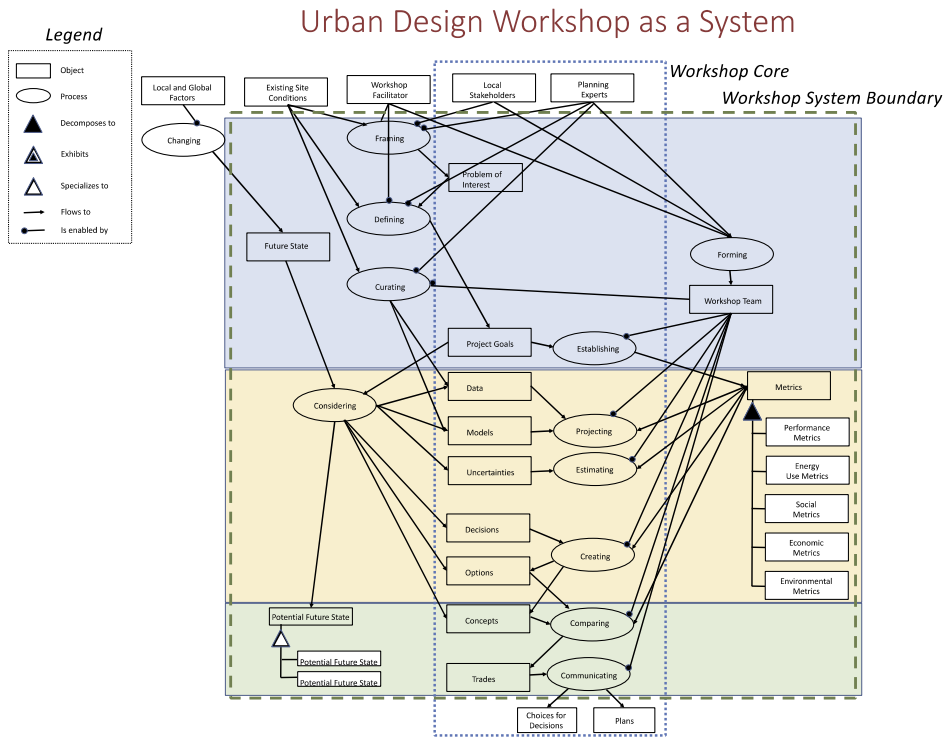


Figure 1. Proposed model for urban design workshop as a system.

A summary of the key processes includes:

- **Framing:** The facilitator convenes the workshop by framing the problem of interest.
- **Curating:** The facilitator curates the workshop team by recruiting and inviting the local stakeholders and planning experts as participants.
- **Considering:** An initial step is to select and agree on the scope of the issues considered in the workshop. From this output, a preliminary agreement is made on the project goals and project processes. Historical knowledge may be incorporated and communicated between participants. Stakeholders may be expected to have local historical knowledge and lived experience historical knowledge. Experts may be expected to have professional historical knowledge about the site of interest, building practices, technical successes and failures, and detailed historical context.
- **Creating, Comparing, and Communicating:** The workshop outputs are expected to yield a range of design options and plans for the ongoing involvement of the participants in future iterations and project progression.

2.2. Research Questions

The research question examines the role for “pre-digested” materials presented by experts. To what extent do interactive models provide for reframing questions? Does this reframing contribute to traceability of ideas from problem to solution? Are there measurable changes in the solutions proposed? [32] [33]. The purpose of this study is to examine a framework considering the transdisciplinary nature for data-driven decisions in urban design and sustainability.

3. Methods and Experiment Design

An experiment is proposed based on the structure of the workshop process presented in this paper. Figure 2 summarizes the proposed workshop experiment sequence. The experiment comprises:

- decision-making by the groups is performed using a computer-based simulation
- empirical measurements are made by monitoring the participants with decisions logged and timestamped through the computer interactions.

The simulation models the site so the participants can explore, analyze, and decide about the solution space. With the concerns defined for the problem space, the participants work in assigned groups to explore building size and land use at the site of interest.

3.1. Workshop Sequence

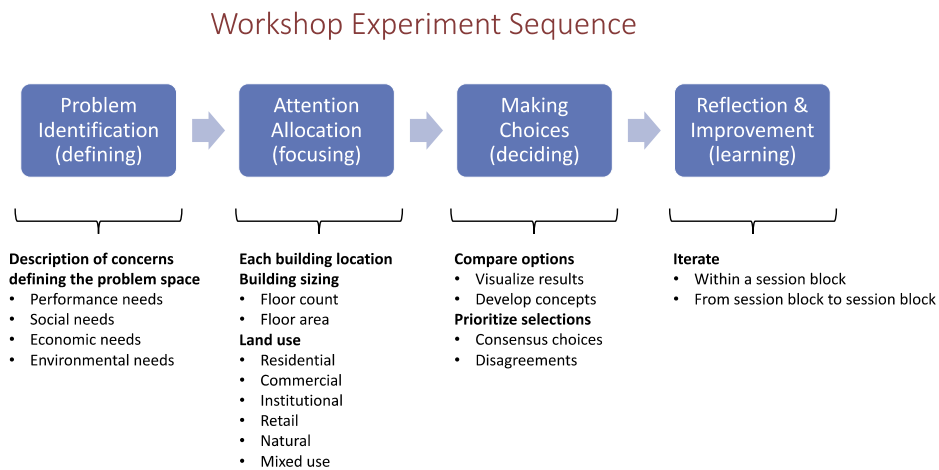


Figure 2. Workshop experiment sequence. The three steps are problem identification (defining), attention allocation (focusing), and making choices (deciding).

4. Proposed Urban Design Workshop

The site of interest focuses on the selected land use case of Ball Square in Somerville, MA. The Massachusetts Bay Transportation Authority (MBTA) identified the area surrounding the newly opened transit station as suitable for transit-oriented development (TOD) within the current expansion program of the MBTA Green Line Extension. The light rail station is one of six new stations and one relocated station first proposed as part of the Central Artery/Tunnel project for Boston-area transportation. The Green Line Extension Review Final Report notes that substantial transit-oriented planning was completed for the Green Line Extension (GLX) and indicates an expectation of increased ridership within a ten-minute walking perimeter at each station, along with improved local air quality to bring economic benefits [34].

4.1. Physical Geography of Somerville, MA

The physical geography presented in Figure 3 maps the land use of the selected site. Social and equity trends highlight the low rate of homeownership compared with local and national percentages as presented in Figure 4 and the changing use of commuter transportation as described in Figure 5.

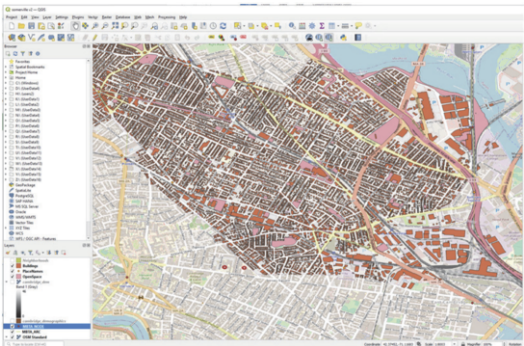


Figure 3. OGIS representation of Somerville, MA. Data sources from Somerville GIS portal, OSM Street Maps, and MBTA/Mass.gov. Legend: Buildings and Structures (Orange and Brown); Open Space and Parks (Pink); Streets (White); Thoroughfares (Yellow). [35], [36].

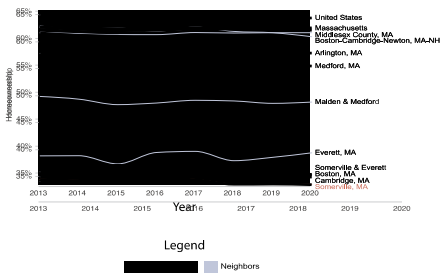


Figure 4. Homeownership for households in Somerville, MA, compared with local and national percentages. (DataUSA.io, US Census Bureau ACS) [37] [38].

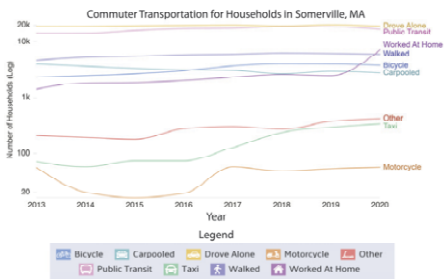


Figure 5. Commuter transportation for households in Somerville, MA. (DataUSA.io, US Census Bureau ACS) [37] [38].

4.2. Site Model Structure

The site of interest is the Ball Square area of Somerville, MA. A set of six stakeholders are identified together with eight buildings of interest. These include the Ball Square T station and surroundings as shown in Figure 6. Images of the site buildings and locations are shown in Figure 7. The pooling model for the study design follows Batty as described in the New Science of Cities [39]. The stakeholders and actors include six categories that represent those with power and interest in this neighborhood. Eight buildings and locations related to the potential for transit-oriented development (TOD) in the Ball Square T station area. These are shown in Figure 8.

Table 2. Ball Square site actors, stakeholders, sites, buildings, and locations.

Actors and Stakeholders	Sites, Buildings, and Locations
1. City of Somerville	1. Somerville DPW
2. Residents	2. Trum Field
3. Retail Commercial Owners	3. Tufts Park
4. Developers	4. Curtis Tufts High School
5. Tufts University	5. Tufts University Campus
6. Property Speculators	6. Broadway Restaurants/Shops
	7. Ball Square T Station
	8. Boston Avenue Site

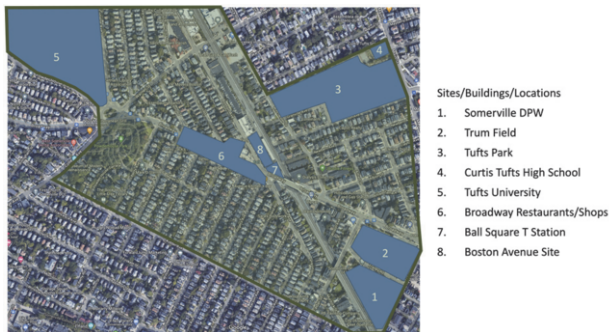


Figure 6. Buildings and locations identified for study at Ball Square, Somerville, MA site at the Green Line Extension T station and surroundings. Data source and images from Google Maps. [35], [36].

Table 1. Attributes of social, economic, and environmental dimensions.

Dimension	Dimension Description and Attributes
Social dimension	• Broadened participation in assessing who responds, how they engage, and evaluating inputs • Quality of Life improvements and trade-offs • Safety and well-being: public safety and health means/ measures • Informed and facilitated participation benefits to social fabric development initial phase and continuum
Economic dimension	• Land usage: planned, flexible-use allocations and nature-based • Resource accounting (natural and man-made): demolish, vacate, improve, adapt, or add • Distributed stakeholder benefits and risks: planning and development industry and community • Invested ownership: existing and future stakeholders • Community stability: short and long-term financial means / measures
Environmental dimension	• Energy consumption, carbon-loop and their relationship to land use management and maintenance • Minimize and reduce carbon footprint • Optimize net embodied carbon • Reduce greenhouse gas emissions • Reduce air pollutant emissions • Avoid unsuitable development • Assess climate change vulnerability • Evaluate risk and resilience

5. Discussion and Interpretation

In a similar pattern related to the three-stage process model presented in this paper, the upstream practice of design may be explored with the downstream disciplines for maintenance and operations. Teamwork studies address complex human interactions and explore the form of design workshops for transdisciplinary engineering to sustainable city planning and development.

Applying this methodology of teamwork research for land use, urban planning, and design offers to realize the systems approach outlined to consider whole systems in context for an organized approach to changes in cities[40]. Future workshop studies offer an approach to test these ideas in the context of urban land-use, environmental dimensions, and sustainable development.

References

- [1] M.M. Sharif, *A Sustainable Urban Future Through Inclusive and Effective Multilateralism: Achieving the Sustainable Development Goals in Times of Global Crises*, UN-HABITAT, Ex. Director's Thematic Paper, Jun. 2023. Accessed: Jun. 07, 2023. <https://unhabitat.org/executive-directors-thematic-paper>
- [2] J. Fokdal, O. Bina, P. Chiles, L. Ojamäe, and K. Paadam, Setting the stage, in *Enabling the City*, Routledge, 2021, pp. 3–15.
- [3] P. Ballering, *Public Participation Definitely Can Add Value To Transit-Oriented Developments: A study of public participation within Dutch TOD practice*, Master's Thesis, Radboud University, 2022. Accessed: Jan. 07, 2023. <https://theses.uibn.ru.nl/handle/123456789/13499>.
- [4] S. Bell, R. Lee, D. Fitzpatrick, and S. Mahtani, Co-producing a Community University Knowledge Strategy, *Frontiers in Sustainability*, 2021, Vol. 2, 661572 ,Accessed: May 27, 2023. <https://www.frontiersin.org/articles/10.3389/frsus.2021.661572>.
- [5] D. Armitage et al., Applying a 'theory of change' process to facilitate transdisciplinary sustainability education, *Ecology and Society*, 2019, Vol. 24, no. 3, doi: 10.5751/ES-11121-240320.
- [6] S. Bartlett and D. Satterthwaite, Eds., *Cities on a Finite Planet: Towards transformative responses to climate change*. Routledge, London, 2016. doi: 10.4324/9781315645421.
- [7] S. Bell et al., Establishing a statement of principles for community engagement with civil engineering, *Proceedings of the Institution of Civil Engineers-Civil Engineering*, Thomas Telford Ltd, 2022, pp. 133–140.
- [8] A. V. Woodworth, D. Purohit, H. Gibson, K. Cressall, R. Desai, and T. Lee, Urban Design Workshop: Third Scale. A Quarter More: Exploring Density in Boston, *The Urban Design Workshop*, 2022. <https://www.architects.org/programs/urban-design-workshop>.
- [9] A. Kennedy, *Participatory design processes: building an evaluation framework*, PhD Thesis, Strathclyde University, 2022, Available: <https://stax.strath.ac.uk/concern/theses/sn009z32q>.
- [10] P. S. Tan and B. R. Moser, Detection of Teamwork Behavior as Meaningful Exploration of Tradespace During Project Design, *Complex Systems Design & Management Asia*, M. A. Cardin, D. Hastings, P. Jackson, D. Krob, P. C. Lui, and G. Schmitt, Eds., in *Advances in Intelligent Systems and Computing*. Springer International Publishing, Cham, 2019, pp. 73–87. doi: 10.1007/978-3-030-02886-2_7.
- [11] R. Basu and J. Ferreira, A LUTI microsimulation framework to evaluate long-term impacts of automated mobility on the choice of housing-mobility bundles, *Environment and Planning B: Urban Analytics and City Science*, 2020, Vol. 47, no. 8, pp. 1397–1417.
- [12] A. Sevtsuk, Estimating Pedestrian Flows on Street Networks, *Journal of the American Planning Association*, 2021, Vol. 87, no. 4, pp. 512–526, doi: 10.1080/01944363.2020.1864758.
- [13] J. Speck, *Walkable city: How downtown can save America, one step at a time*. Macmillan, New York, 2013.
- [14] J. Speck, *Walkable city rules: 101 steps to making better places*. Island Press, Washington, 2018.
- [15] K. Kimura, *The Effect of Introducing New Transportation Services on the Community Engagement of Elderly People and Parents*, Thesis, Massachusetts Institute of Technology, 2021. Accessed: Jan. 30, 2023. <https://dspace.mit.edu/handle/1721.1/138918>.
- [16] A. Jain, D. Robinson, B. Dilkina, and R. Fujimoto, An approach to integrate inter-dependent simulations using HLA with applications to sustainable urban development, *2016 Winter Simulation Conference (WSC)*, IEEE, 2016, pp. 1218–1229.

- [17] R. Basu, J. Ferreira, and R. Ponce-Lopez, A framework to generate virtual cities as sandboxes for land use-transport interaction models, *Journal of Transport and Land Use*, 2021, Vol. 14, no. 1, pp. 303–323.
- [18] B. Moser, C. Bellew, A. Cadario, and R. Filippi, The Interplay of Site and Construction Systems Decisions for Complex Urban Infrastructures, *Advances in Transdisciplinary Engineering*, 2019, Vol. 10, pp. 497–508, doi: 10.3233/ATDE190157.
- [19] K. E. Raynor, A. Doyon, and T. Beer, Collaborative planning, transitions management and design thinking: evaluating three participatory approaches to urban planning, *Australian Planner*, 2017, Vol. 54, no. 4, pp. 215–224.
- [20] E. Bruch and F. Feinberg, Decision-Making Processes in Social Contexts, *Annu Rev Sociol*, Jul. 2017, Vol. 43, pp. 207–227, doi: 10.1146/annurev-soc-060116-053622.
- [21] M. R. Endsley, Toward a Theory of Situation Awareness in Dynamic Systems, *Human Factors*, 1995, Vol. 37, No. 1, pp. 32–64.
- [22] L. Pelegrin, B. Moser, and V. Sakhiani, A Quasi-Experiment to Expose Attention-Decision-Learning Cycles in Engineering Project Teams, *Proceedings of the 52nd Hawaii International Conference on System Science*, 2019, <https://hdl.handle.net/10125/59475>
- [23] P. Manandhar et al., Sensing systemic awareness and performance of teams during model-based site design, *2020 IEEE 6th World Forum on Internet of Things (WF-IoT)*, IEEE, 2020, pp. 1–6.
- [24] I. Vazquez, F. Faruque, and B. Moser, Instrumenting Weick’s Seven Properties to Measure Collective Sensemaking in Engineering Teams, *Advances in Transdisciplinary Engineering*, 2022, Vol. 28, pp. 289–298.
- [25] K. P. McDonough, *Detecting the influence of stakeholders’ mental models on emergent collective awareness in instrumented teamwork workshops*, Master’s Thesis, Massachusetts Institute of Technology, 2021. <https://dspace.mit.edu/handle/1721.1/139316>
- [26] S.W. Kozlowski and D.R. Ilgen, Enhancing the effectiveness of work groups and teams, *Psychological science in the public interest*, 2006, Vol. 7, no. 3, pp. 77–124.
- [27] S. W. Kozlowski, Enhancing the effectiveness of work groups and teams: A reflection, *Perspectives on Psychological Science*, 2018, Vol. 13, no. 2, pp. 205–212.
- [28] C. Ataman and B. Tuncer, Urban Interventions and Participation Tools in Urban Design Processes: A Systematic Review and Thematic Analysis (1995–2021), *Sustainable Cities and Society*, 2022, Vol. 76, 103462.
- [29] M. Kopáček, Land-Use Planning and the Public: Is There an Optimal Degree of Civic Participation?, *Land*, 2021, Vol. 10, no. 1, 90.
- [30] J.A. Todd and G. Lindsey, *Planning and Conducting Integrated Design (ID) Charrettes*, Nov. 19, 2016. <https://www.wbdg.org/resources/planning-and-conducting-integrated-design-id-charrettes> (accessed Jan. 17, 2023).
- [31] G. Lindsey, J. A. Todd, S. J. Hayter, and P. G. Ellis, *A Handbook for Planning and Conducting Charrettes for High-Performance Projects*, 2nd ed., National Renewable Energy Laboratory, Golden, Colorado, NREL/BK-550-44051, Sep. 2009.
- [32] S. E. Bibri, The core academic and scientific disciplines underlying data-driven smart sustainable urbanism: an interdisciplinary and transdisciplinary framework, *Comput. Urban Sci.*, Mar. 2021, vol. 1, no. 1, p. 1, doi: 10.1007/s43762-021-00001-2.
- [33] C. J. Duerksen, C. G. Dale, and D. L. Elliott, *The Citizen’s Guide to Planning*, 4th edition. American Planning Association, Chicago, 2009.
- [34] {GLX Constructors}, GLX Constructors: Ball Square Station Information and Design – January 30, 2019 - Google Search, Jan. 2019. Accessed: Dec. 28, 2022. <https://www.mass.gov/doc/ball-square-station-design>
- [35] City of Somerville, *City of Somerville, MA GIS Data*, City of Somerville, MA, 2022. <https://data.somervillema.gov/browse?category=GIS+data> (accessed Nov. 22, 2022).
- [36] MASS.gov, *MassGIS Data: MBTA Rapid Transit*, Dec. 2021. <https://www.mass.gov/info-details/massgis-data-mbta-rapid-transit> (accessed Nov. 22, 2022).
- [37] DataUSA.io, DataUSA.io. Datawheel, 2023. Accessed: Jan. 27, 2023. <https://datausa.io/>
- [38] US Census Bureau, American Community Survey (ACS), *Census.gov*, 2022. <https://www.census.gov/programs-surveys/acs> (accessed Dec. 15, 2022).
- [39] M. Batty, *The New Science of Cities*. The MIT Press, 2017. Accessed: Jan. 14, 2023. <https://mitpress.mit.edu/9780262534567/the-new-science-of-cities/>
- [40] A. Harding, B. Hecht, and H. Love, Design for Action: Networking with Global Technical Societies for Sustainable and Inclusive Systems, *United Nations 68th Civil Societies Conference*, Salt Lake City, Utah, 2019. Accessed: Jan. 12, 2023. <https://www.un.org/en/68th-united-nations-civil-society-conference>